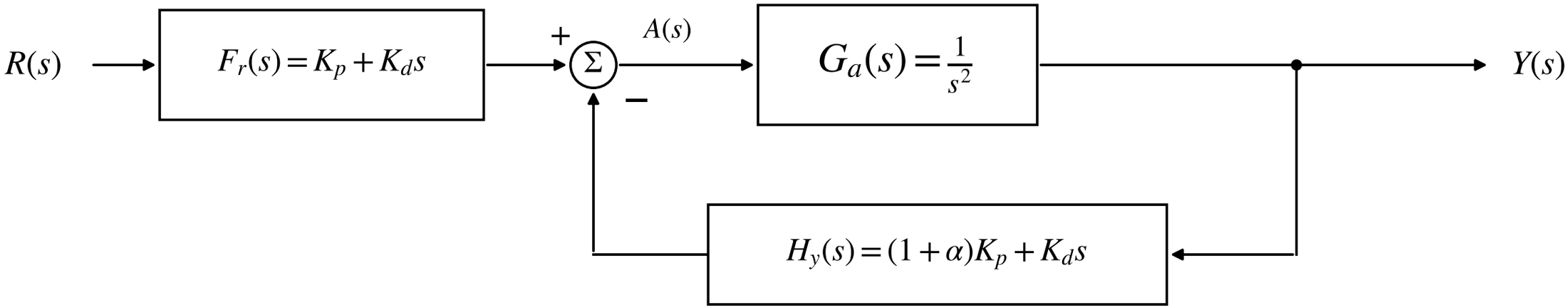
V16 | Continuous-time transfer-function block diagram

Ideal compensated joint axis: r = q_ref - q0, y = q - q0. The sampled equivalent is on the next appendix page.



$$\frac{Y(s)}{R(s)} = \frac{K_d s + K_p}{s^2 + K_d s + (1+\alpha)K_p}$$

The underlying torque-to-angle plant retains the 12-joint coupling:

$$G_{\tau q}(s) = \left(M_0 s^2 + D_0 s + K_g\right)^{-1} \in \mathbb{C}^{12\times 12}$$

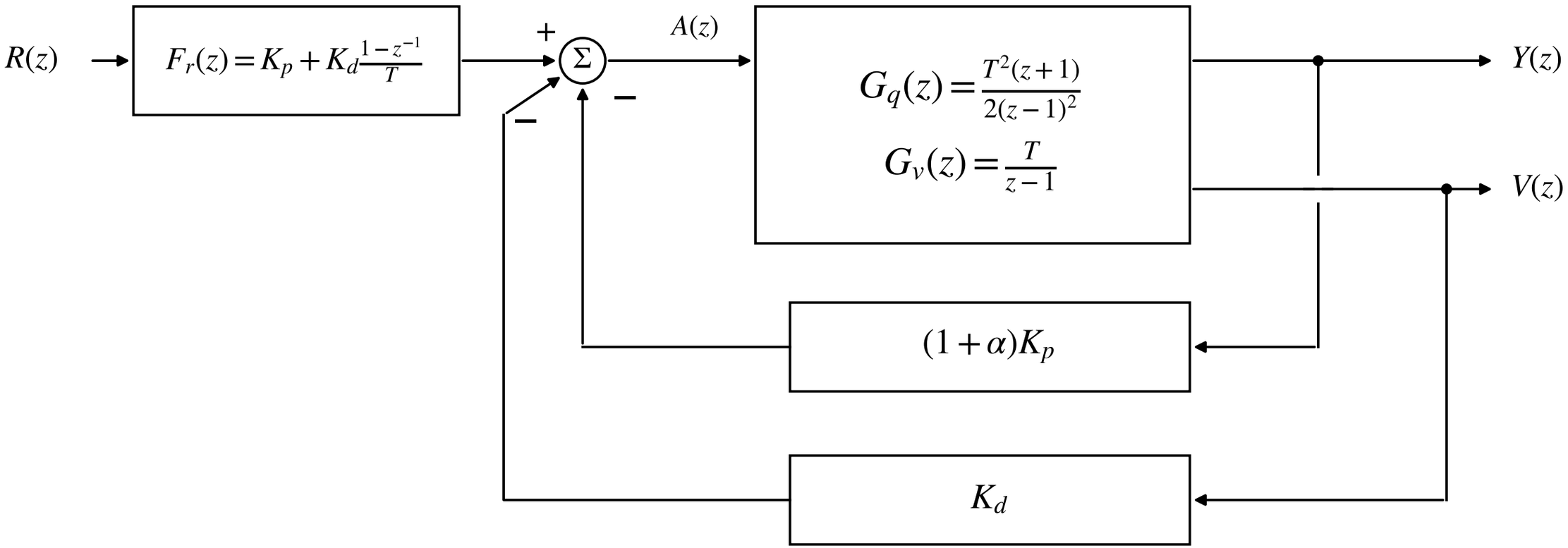
$$Y(s) = G_{\tau q}(s)U(s) + Y_{\mathrm{IC}}(s), \quad Y_{\mathrm{IC}} = G_{\tau q}[M_0(sy_0 + v_0) + D_0 y_0]$$

$$G_{\tau q,\,d}(z) = C_q(zI - A_d)^{-1}B_d, \qquad (A_d, B_d) = \mathrm{ZOH}_{\tau,T}(A, B)$$

Upper loop: continuous-time equivalent of the joint acceleration feedback, with ideal inverse dynamics.
Lower plant: fixed-base, no-contact source nominal model; D0 = 1.5 I. Actual USD damping is 0.
Transfer functions describe zero-initial input response; the initial-condition source is shown separately.

V16 | z-domain feedback: ideal acceleration-ZOH channel

T = 0.020 s. Ideal acceleration hold after compensation; actual V16 holds torque. Measured velocity is a separate feedback.



$$\frac{Y(z)}{R(z)} = \frac{G_q(z)F_r(z)}{1 + (1+\alpha)K_p G_q(z) + K_d G_v(z)}$$

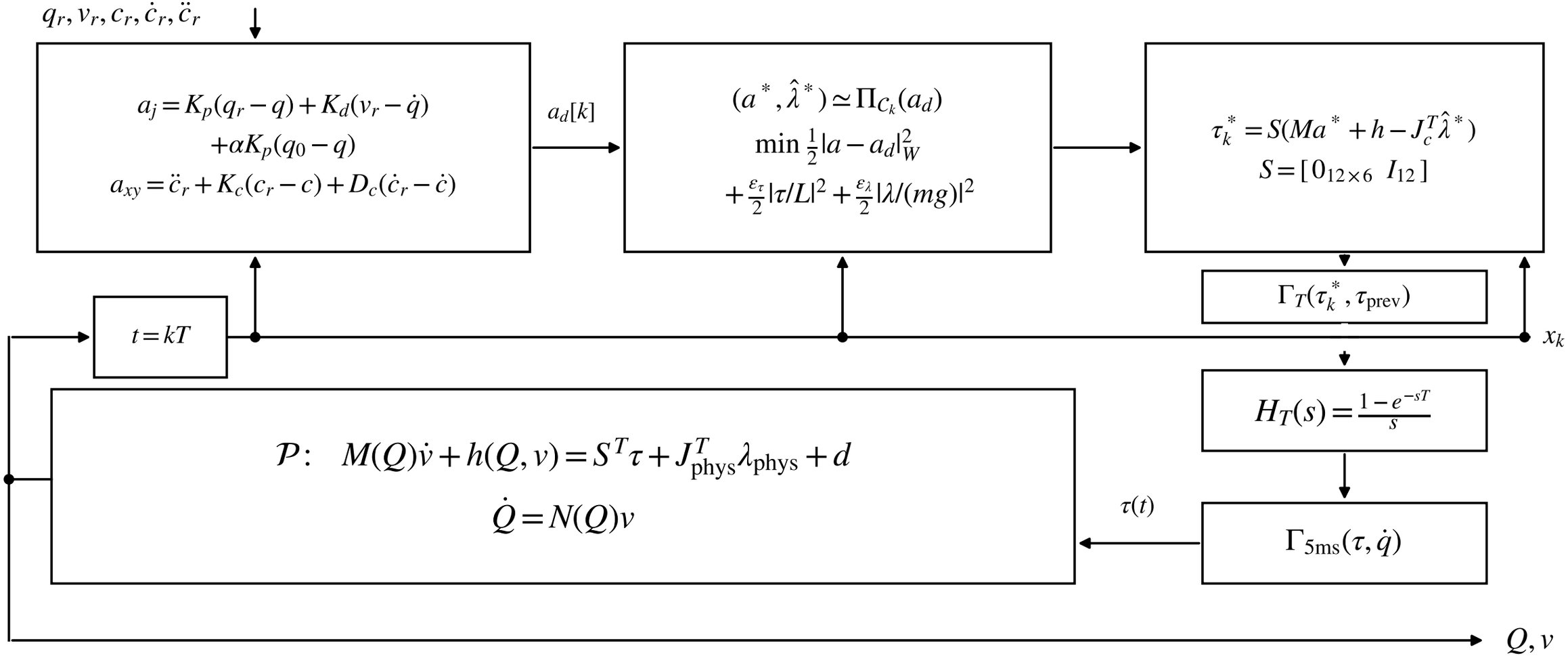
$$D_{\mathrm{cl}}(z) = (z-1)^2 + K_d T(z-1) + \tfrac{1}{2}(1+\alpha)K_p T^2(z+1)$$

Contact QP / 571: Kp = 319.305 s^-2, Kd = 27.468 s^-1, alpha = 0.529243
Whole-body QP / 577: Kp = 289.332 s^-2, Kd = 27.173 s^-1, alpha = 0.647501

Dcl is the homogeneous feedback characteristic; Fr adds reference memory z^-1. Reference velocity / acceleration clipping and QP changes are inactive.

V16 | Actual nonlinear sampled control law

Mathematical blocks for the trained whole-body QP controller; Q has 19 coordinates, v has 18 components, tau has 12.



$$C_k:\ M_b a + h_b = J_{c,b}^T\lambda, \quad J_c a = a_c, \quad \tau_{\min} \le \tau \le \tau_{\max}$$

$$0 \le \lambda_z \le 500, \quad |\lambda_x|, |\lambda_y| \le \frac{0.6}{\sqrt{2}}\lambda_z, \quad \lambda_{\text{inactive}} = 0$$

IK/reference feedback and acceleration clipping are inside the sampled control law. QP uses 320 finite ADMM iterations.
Gamma_T: 20 ms slew clipping before hold. Gamma_5ms: speed/effort envelope after hold. Model and physical contact forces are distinct.